

Response to Reviewers

Manuscript: *Proxy-Based Diagnostics of Quantum Oracle Sketching Robustness for Non-IID Sensor and Telemetry Streams*

Journal: *Sensors* (MDPI) · **Manuscript ID:** sensors-4470240

Dear Editor,

Thank you for the opportunity to submit a revised version of our manuscript, *Proxy-Based Diagnostics of Quantum Oracle Sketching Robustness for Non-IID Sensor and Telemetry Streams* (Manuscript ID: sensors-4470240), to *Sensors*. We appreciate the time and effort that you and the reviewers have dedicated to the paper, and we are grateful for the constructive comments, which have improved it considerably.

Both reviews converge on the same central point, and we agree with it. The previous version, although hedged internally, still read as evidence about quantum performance, whereas what the work can honestly support is a proxy-based diagnostic and hypothesis-generation framework for correlation-sensitive streaming analysis. The revision repositions the manuscript accordingly, in the title, abstract, statistics, figures and conclusions, without removing any experiment or altering any measured result. All stochastic outputs regenerate identically from the same seeded pipeline, which we verified before beginning the re-analysis. In accordance with the Academic Editor’s request that preprints not be cited, the Zhao et al. (2026) preprint has been removed from the manuscript and the reference list, which now contains 54 references; no other work was substituted for it, and the affected statements were revised so that each remaining reference supports only what it actually establishes.

The changes are marked in the highlighted version submitted with this letter. A point-by-point response to the comments follows.

Comments from Reviewer 1

Comment 1: *I strongly recommend to reduce the emphasis on quantum advantage and instead presenting the τ, r landscape primarily as a diagnostic framework to identify scenarios where future quantum implementations is most promising.*

Response: Thank you for this recommendation, which we have adopted as the organising principle of the revision. The manuscript is now presented as a diagnostic and hypothesis-generation framework rather than as evidence of quantum advantage. The title now leads with “Proxy-Based Diagnostics”; the Introduction states that the paper “neither demonstrates nor claims a realised quantum-classical separation”; and the Conclusions identify the (τ, r) landscape as the primary contribution, with hypothesis generation as the secondary one. Section 7.4 states three falsifiable hypotheses, H1 to H3, each with an explicit refutation criterion, so that the question of where future implementations are most promising becomes concrete and testable. Throughout the paper, “quantum-favourable” has been replaced by “proxy-favourable” or “hypothesized favourable”.

Comment 2: *Please clarify the relationship between the heuristic proxy model and the original theoretical framework.*

Response: Thank you for raising this. A dedicated subsection now draws that boundary explicitly. Section 3.3 separates five things: the published quantum-streaming results that provide the general context, together with the statement that these are problem-specific and do not establish the operational τ, r, T_{eff} or accuracy model used here; the assumptions this paper adopts; the heuristic constructions it introduces, namely

the effective-sample correction in T_{eff} and the VC-type task rate; the memory curves, retained only as illustrative scaling references; and the scope limitations. Table 1 records the provenance and epistemic status of every model-level quantity, with a column stating what is not theoretically guaranteed. The relationship is therefore presented as one of motivation and analogy rather than derivation, and the manuscript states that it adds no quantum theorem.

Comment 3: *The analytic Markov τ proxy differs substantially from the empirical measurements. I recommend to discuss this discrepancy in greater detail and explain whether it reflects limitations of the proxy model or the synthetic data generation process.*

Response: We agree that this deserved fuller treatment, and Section 7.3 is now devoted to it. For Markov switching the two quantities answer different questions: each bit’s marginal autocorrelation decays as ρ^k and crosses $1/e$ at about 2.8 lags at $\rho = 0.7$, whereas the joint n -bit chain requires on the order of $n \log n / (1 - \rho)$ steps to mix. Both values are correct answers to different questions, so the divergence is partly expected behaviour and partly a limitation of summarising dependence in a single scalar. It is not an artefact of the data generation: the generators reproduce their intended statistics, and the gap persists under all four alternative tau estimators examined in Experiment 9, which isolates the estimator family as the cause. The subsection also sets out the practical consequences, namely that the empirical tau serves as the operational input, the analytic tau as a conservative envelope, and their ratio as a diagnostic for long-memory content.

Comment 4: *I recommend to validate the proposed framework on additional real-world datasets. It will help to better demonstrate its generalizability.*

Response: We agree that this is the most valuable extension, and we are explicit that it has not been carried out in the present revision. Adding datasets would have required a new experimental campaign, and we preferred to keep this revision re-analysis based so that every published number remains reproducible from the pinned seeds. What we have added, from the same raw data, is an encoder and resolution sensitivity analysis in Section 6.8: the two streams were re-encoded across ten combinations of sampling resolution and bucket width, and the ordering of NYC Taxi above Machine Temperature holds in all ten. The reported placement is therefore not an artefact of one encoder setting. The manuscript now states that the real-stream experiment is illustrative rather than statistically representative, the Threats section records the two-dataset limitation, and Future Work item (iii) commits to network telemetry, cybersecurity event streams, financial series and IoT fleets, with hypothesis H1 defining the protocol by which generalisability will be tested.

Comment 5: *Please include a more comprehensive sensitivity analysis of the selected proxy parameters, particularly C and δ , by this you will show how they influence the reported results.*

Response: Agree, and we have added this quantitatively. Because the proxy is a closed-form function of the measured tau and r , it can be re-evaluated on the same per-seed values across C in $\{1, 1.5, 2, 2.5\}$ crossed with δ in $\{0.01, 0.05, 0.1\}$. The new Table 8 reports the resulting movement of the proxy-baseline crossover. The ordering of the regimes is invariant, since C and δ enter only through the monotone factor $C^2 \log(1/\delta)$, but the crossover location moves from about $\rho^* = 0.38$ at $C = 2.5$ and $\delta = 0.01$ to beyond the sweep range at $C = 1$, with ρ^* about 0.82 at the operating point. The text now instructs that the landscape be read ordinally, the abstract carries this conditionality, and Section 5.3 states that $C = 2$ is a conventional order-unity choice not fixed by any constant in the cited literature.

Comment 6: *The comparison between deterministic quantum proxies and stochastic classical baselines should be presented more cautiously. The current statistical analysis may unintentionally suggest a comparison between implemented algorithms rather than between empirical results and theoretical proxies.*

Response: This is an important point and we have rebuilt the statistical presentation around it. Table 4 now reports Hodges-Lehmann estimates of the paired difference with exact signed-rank 95% intervals and

matched-pairs rank-biserial correlations, and the p-values are retained only as descriptive companions. A dedicated paragraph in Section 6.5 explains that the pairing is between a stochastic learner and a model prediction, that the proxy’s seed-to-seed variation comes solely from the per-seed tau estimate, and that the rank-biserial value here indexes sign concordance across seeds rather than magnitude. The same paragraph states that the tests “do not, and cannot, establish that one algorithm outperforms another, and no quantum algorithm is implemented anywhere in this comparison”. We also disclose that none of the five comparisons survives Holm correction at $n = 10$, that the interval at $\rho = 0.48$ is not robust to a single step of the integer-valued tau estimator, and that the crossover location is a deterministic solution whose uncertainty derives from the estimator and the constants rather than from the tests.

Comment 7: *I recommend to provide a stronger justification for the selected classical baselines and discussing if additional streaming methods can offer a more informative comparison.*

Response: Thank you for this suggestion. Section 5.2 now justifies the choice explicitly. The three baselines span complementary axes of bounded-memory streaming learning, namely gradient-based (online SGD), variance-reduced gradient-based (averaged SGD) and non-gradient hash-count-based (Count-Min), while sharing the single-pass, fixed-memory regime that the proxy models. We also explain why drift-adaptive methods such as ADWIN-equipped learners and adaptive random forests are deliberately excluded from the primary comparison: they respond to non-stationarity by discarding history, which would confound the correlation-budget question the proxy isolates. Benchmarking them on the same landscape is named as an extension in Future Work.

Comment 8: *Please explain more clearly why the proposed operational proxies, τ and r , are sufficient to characterize the correlation properties of the studied data streams, and discuss any limitations of relying only on these two measures.*

Response: We agree, and the revision now treats two-scalar sufficiency as an assumption rather than a fact. Section 3.3 lists it under Assumptions as a modelling choice of this paper, and Section 7.3 shows its failure mode: for non-mixing, long-range-dependent processes there is no single scalar on which the analytic and empirical estimators could agree, so a single-number tau is intrinsically lossy in that regime. Two mitigations are stated. The ratio of analytic to empirical tau is itself a useful third diagnostic that flags long-memory content, and multi-scale refinements of tau, using integrated-autocorrelation estimates at several horizons, are named as follow-on work. The Threats section records the residual risk under construct validity.

Comment 9: *I recommend to expand the discussion of the assumptions behind each synthetic correlation regime and to please explain how closely these settings represent practical sensor and telemetry environments.*

Response: Thank you for this suggestion. Section 4 already carried per-regime provenance notes, and the summary in Section 4.6 now adds the realism mapping. Markov switching idealises short-range state persistence in sensor readouts; seasonal drift represents the diurnal and weekly cycles of telemetry and demand series; burst repetition represents the duplicate-heavy event bursts of monitoring and logging pipelines; and long-range dependence represents the self-similar behaviour first measured on Ethernet traffic. We state plainly that these are idealisations, that real streams mix the mechanisms, as the two NAB cases show, and that the mapping to any concrete deployment is qualitative.

Comment 10: *The manuscript would benefit from a dedicated discussion of threats to validity, include the assumptions introduced by the heuristic proxy model, synthetic data generation and feature binarization.*

Response: Agree. This is now a dedicated section. Section 8, “Threats to Validity”, is organised by construct, internal, external and statistical-conclusion validity, and covers the items you list. It addresses the

heuristic proxy constructs, with pointers to the provenance table; the selection of synthetic generators; feature binarisation, noting that arcsine shrinkage makes the real-stream τ and r estimates lower bounds on the correlation of the underlying continuous process; the design of the deterministic-versus-stochastic comparison; dataset selection; the definition of the target, including the full-series threshold look-ahead raised in review; the fixed hyperparameters; and the ten-seed budget.

Comment 11: *I recommend to discuss the computational complexity of estimating τ and r and the scalability of the proposed framework for high-volume streaming systems.*

Response: Thank you for pointing this out. Section 5.1 now contains a paragraph on estimator cost and scalability. The τ estimator requires the bitwise autocorrelation up to lag K , which costs $O(nTK)$ time and $O(nK)$ memory using a ring buffer per feature. The r estimator scans a forward window of length W , costing $O(TW)$ comparisons, or expected $O(T)$ with hashing, and $O(Wn)$ memory. Both are linear in T , independent of the alphabet size, single-pass friendly and parallel across features. On the longest stream studied, $T = 22,695$, they complete in well under a second, so the diagnostic adds negligible overhead in high-volume deployments.

Comment 12: *Some figures contain many overlapping curves and annotations, I recommend to simplify the visual presentation and enlarge labels to improve readability.*

Response: Thank you for this observation. All nine figures were re-audited and several were regenerated. The legend entry “Proxy-model advantage” in Figure 5 has been renamed “Proxy-estimated separation”, and stray typographic characters were removed from five panel titles. In-image labels that referred to qubit counts and to quantum order-notation were replaced with neutral proxy labels in Figures 1, 2 and 6. The synthetic markers in Figure 8(d) are now identified as schematic orientation marks, and the caption gives the numeric stream coordinates instead of relying on visual adjacency. Confidence-interval statements in captions were scoped to the stochastic panels only, and four tables that previously overran the text width were reformatted. All regenerations are rendering changes only; the underlying numerical outputs are unchanged.

Comment 13: *The discussion could better distinguish theoretical quantum memory advantages from the practical observations obtained through the proxy-based evaluation.*

Response: We agree, and this distinction is now enforced wherever the memory curves appear. Experiment 6 opens with the statement that no memory lower bound is measured empirically in that experiment and that no result constitutes an empirical memory floor for online SGD, averaged SGD or Count-Min. Section 3.3, under “Memory references”, records that the published quantum-streaming papers do not establish the $O(n)$ and $\Omega(\sqrt{N})$ pair for the diagnostic task studied here, and that the pair is retained solely as a paper-defined illustrative scaling reference. Table 1 lists that pair as “introduced here (illustrative scaling reference)”, with “a proven separation for this diagnostic task” in the not-guaranteed column. The caption of Figure 6 labels panel (a) as an illustrative memory-scaling reference with no measured content, and Section 7.1 closes its three structural factors with the statement that all three are properties of the proxy model or its illustrative references and that none is a measured quantum result. The statement about $n \geq 20$ is marked as an extrapolation beyond the tested range.

Comment 14: *Please discuss the applicability of the proposed framework to other streaming domains beyond sensor telemetry, such as network traffic, financial streams, or cybersecurity monitoring.*

Response: Thank you for this suggestion. Table 9 now maps network backbone traffic, financial returns, sensor and IoT streams, and concept-drifted streams onto the landscape using published measurement ranges, with the values recomputed from the paper’s own formula, the verdicts given as ranges, and the estimator convention stated. Future Work item (iii) commits to the domains you name: network telemetry in the form

of flow and latency series, cybersecurity event streams such as intrusion and log-anomaly feeds, financial tick and returns series, and large-scale IoT monitoring fleets.

Comment 15: *strengthen the conclusion, for this emphasize the practical contributions of the study, clearly summarize its limitations, and please outline the most important directions for future research.*

Response: Agree. The Conclusions have been rewritten to this structure. A paragraph on the practical contribution presents the landscape as a low-cost pre-screening step: estimate τ and r , place the stream, and read off whether correlation alone erodes the effective budget below the level needed for rare-event learning, as it does for the Machine Temperature stream at $T_{\text{eff}} = 52$. A second paragraph on limitations and required validation points to the Threats section and names the decisive missing step, namely an implemented or simulated sketching pipeline evaluated at coordinates the landscape predicts to be favourable and unfavourable. The research-question answers are restated in effect-size terms with the estimator convention made explicit in each case.

Comments from Reviewer 2

Comment 1: *The key issue is the relationship between τ and r and the quantities in the framework of quantum oracle-sketching. ... They have to strengthen the theoretical basis for the chosen proxy construction and distinguish the results from the original from the assumptions that are used in this paper.*

Response: Thank you for identifying this as the key issue. Section 3.3 now addresses it directly and states the boundary rather than assuming an equivalence. The section has five labelled parts. The first sets out the published quantum-streaming results that motivate the study, namely a polynomial quantum space advantage for one-pass triangle counting, an exponential quantum space advantage for approximating maximum directed cut, and a black-box route from certain classical sketches to quantum streaming algorithms, together with the statement that these are problem-specific results that do not establish the operational τ , r , T_{eff} or accuracy model used here and that this paper adds no quantum theorem. The remaining parts cover the assumptions this paper adopts, the heuristic constructions it introduces, the memory curves retained as illustrative scaling references, and the scope limitations, which state that τ and r are measurement conventions and are not formally equivalent to a universal mixing time, an effective-sample count or a quantum sample-overhead parameter. Table 1 gives the provenance of each quantity with an explicit not-guaranteed column. In direct answer to your question, τ and r are operational summaries defined in this paper for the proxy-based diagnostic; the manuscript neither implements the framework nor claims to prove or reproduce any quantum result.

Comment 2: *The terminology of quantum performance should be moderated throughout the paper. ... we will refer to them as proxy-based predictions instead of the actual quantum performance. This is particularly relevant in the Abstract, Results, Discussion, figure captions and Conclusion.*

Response: Agree, and a systematic terminology pass was made over each of the surfaces you list. The term “quantum-favourable” no longer appears and has been replaced by “proxy-favourable” or “hypothesized favourable”. Constructions such as “quantum performance proxy crosses below classical performance” have been rewritten, for example as “the accuracy proxy computed with the analytic τ lies below the measured classical baseline”. The relevant column of Table 6 is now headed “Proxy hypothesis”. Because captions alone cannot correct text that is part of an image, Figures 1, 2 and 6 were regenerated to remove in-image labels referring to qubit counts and quantum order-notation. Statements that no quantum algorithm is implemented now appear in the Abstract, in Section 3.3, in four results subsections, in the Threats section and in the Conclusions.

Comment 3: *The statistical comparison between classical baselines and the proxy model is still not resolved.*

Our Wilcoxon tests compare stochastic classical results for seeds to a proxy value that is deterministic for a given value of parameters. ... We believe the magnitude of the classical minus proxy difference and the uncertainty may be a better gauge than simply p-values.

Response: We agree, and have adopted this in full, so that magnitude and uncertainty now lead and the tests are re-scoped to what they can support. For each sweep point, Table 4 reports the Hodges-Lehmann estimate of the paired difference with its exact signed-rank 95% interval, for example +0.042 with interval [+0.011, +0.087] at $\rho = 0$ and -0.046 with interval [-0.079, -0.003] at $\rho = 0.88$, together with the rank-biserial correlation, which we interpret as sign concordance across seeds because the comparison is against a near-constant reference and all magnitude information resides in the difference estimate. The p-values are retained only as descriptive companions. We disclose that none survives Holm correction at $n = 10$ and why, and we disclose the sensitivity of the integer-valued tau estimator, where one lag step corresponds to 0.012 in accuracy and is sufficient to move the interval at $\rho = 0.48$ across zero. The interpretation paragraph states that the crossover is the deterministic solution of the equation between the proxy accuracy and the mean classical accuracy, with uncertainty inherited from the estimator and the constants rather than from the tests. The full re-analysis, which first reproduced the published per-seed results to six significant digits, is included with the revision.

Comment 4: *The huge discrepancy between the analytic and empirical refreshing time is one of the most important findings in the paper, especially for the long-range dependence regime. ... The authors should discuss this in more detail and discuss its implications for the interpretation of the central (τ, r) sensitivity landscape.*

Response: We agree that this is one of the paper’s more important findings and have promoted it accordingly. The new Section 7.3 gives the mechanism: the analytic proxies target model-level worst-case scales, that is joint-chain mixing and tail-dominated integrated autocorrelation, whereas the empirical estimator measures the short-lag marginal decorrelation of the binarised stream, and for a non-mixing process there is no scalar on which the two could agree. The consequences for the landscape are carried through the paper. Verdicts are now bound to their estimator convention wherever they appear, and we state openly that the burst and long-memory extremes exchange places between the analytic and empirical conventions, because burst saturates through its measured duplication while the high empirical value for long memory is a short-lag artefact. This appears in Experiment 9, in Section 7.3, in the answers to RQ2 and RQ3, and in the captions of Figures 3 and 7. The abstract now says that the two estimates diverge, rather than that one overestimates the other, and adds that they answer different questions.

Comment 5: *The classical-memory comparison needs to be taken with care. The classical lower bound curve ... is not actually an empirical lower bound for the SGD, averaged-SGD, or Count-Min tools used in our experiments. ... we should make it more prominent in the dimension-scaling discussion.*

Response: Agree, and this is now made prominent in exactly that discussion. The curves are no longer presented as cited bounds at all. Experiment 6 opens with the statement that no memory lower bound is measured empirically in the experiment, that the curves in panel (a) are illustrative quantities, and that no result constitutes an empirical memory floor for online SGD, averaged SGD or Count-Min. The caption of Figure 6 labels panel (a) as an illustrative memory-scaling reference that is paper-defined and contains no measured content, and states that the exponential divergence of the curves is a visual scaling contrast rather than an established separation. Section 3.3 records under “Memory references” that the published quantum-streaming papers do not establish this pair of curves for the diagnostic task studied here and that they are retained solely as a paper-defined illustrative reference. Table 1 carries the same status, and Section 7.1 states that none of its structural factors is a measured quantum result.

Comment 6: *The real-stream analysis is useful as a sanity check but it is limited to two NAB datasets. The*

conclusions on the sensor and telemetry applications should therefore be conservative.

Response: We agree, and the conclusions drawn from the real streams are now explicitly conservative. Section 6.8 states that the experiment is illustrative rather than statistically representative, the external-validity paragraph records that generalisation is unverified, and the abstract claims placement only. In addition, a new encoder and resolution sensitivity table strengthens what can legitimately be said from two streams: across ten re-encodings of the same raw series, using three sampling resolutions and three bucket widths, the ordering of NYC Taxi above Machine Temperature holds in every configuration, so the reported placement is at least stable with respect to the encoder. Broader corpora are committed to in Future Work, with a concrete test protocol given as hypothesis H1.

Comment 7: *the target defined based on a percentile of the full time series should also be reconsidered in the context of streaming analysis, since using the full series to set the target threshold can take into account future observations. A threshold set from a first calibration interval or updated only from previously observed samples would provide a more rigorous prequential design.*

Response: Thank you for this observation, which we consider well taken, and which we now disclose explicitly rather than leave implicit. Section 6.8 states that the threshold is computed from the full series and therefore uses information that would not be available prequentially at each step. It also states that the feature stream and the tau and r estimates involve no such look-ahead, so the landscape placement, which is the central output for these streams, is unaffected, and the leakage touches only the classical baseline reference numbers. We further state that a threshold calibrated on an initial interval, or updated only from previously observed samples, is the more rigorous prequential design, and we adopt it as future work. The Threats section records this as a residual risk under internal validity. We chose disclosure over re-running the baselines in this revision in order to keep every published number reproducible, and we would be glad to re-run them with a calibration-interval threshold in a further round if you prefer.

Comment 8: *The interpretation of the NYC Taxi Count-Min result also needs to be qualified. The six-bit representation gives a relatively small number of possible feature vectors and hence memorising the same feature patterns is relatively easy. ... I would like to give more weight to the implications when we think about the apparent agreement between the Count-Min performance and our proxy prediction.*

Response: Agree, and the qualification now carries weight in three places. The dedicated paragraph explaining why Count-Min performs well is retained, and the interpretation now states plainly that the favourable-side separation between the two streams rests on Count-Min alone and that the hashed linear learners do not corroborate it. We additionally report that both SGD variants fall below the majority baseline on accuracy and that the balanced accuracy of averaged SGD straddles chance, so the apparent agreement between Count-Min and the proxy prediction is explicitly not read as validation. Section 7.4 lists representation-specific effects of this kind among the phenomena the landscape cannot anticipate, and, going further in the direction you indicate, the same section concedes that, read strictly as an accuracy projection, the proxy is falsified on the Machine Temperature stream, where 0.500 is predicted and 0.72 to 0.80 observed, with the rare-class diagnostic reading offered as interpretation rather than confirmation. Table 6 now carries the ordering-to-ordering intervals, with Count-Min F1 at 0.644 plus or minus 0.062 and a per-ordering range of 0.43 to 0.72, which tempers the headline number.

Comment 9: *In particular, the target function ablation should be highlighted because it shows that our classical performance can change dramatically with the classification task in question and that our proxy is in fact target independent. A correlation-based proxy that is insensitive to the target function cannot necessarily predict the performance of algorithms for different kinds of decision problems.*

Response: We agree, and this is now highlighted and stated as a scope boundary in your own terms. Section

6.9 states that classical learner performance is strongly target-dependent while the proxy depends only on τ , r and T , and is target-independent by construction, so that the proxy cannot universally predict downstream classifier performance and instead characterises the correlation structure of the stream, that is the effective sample budget available to any bounded-memory learner, rather than task-specific learnability. Section 7.4 elevates this to the list of things the landscape cannot do, adding the empirical observation that measured classical accuracy is flat while T_{eff} falls fivefold, so the budget model does not predict even the classical arm’s accuracy. The caption of Figure 9 states that the proxy does not track task-specific learnability, and the answer to RQ2 now describes τ and r as budget summaries rather than accuracy predictors. The protocol of the ablation, which is prequential with a learning rate of 0.01 and $T = 4096$, is also stated so that its absolute levels are not misread against the main experiments.

Comment 10: *the strongest contribution of the paper is the (τ, r) sensitivity landscape proposed as a hypothesis-generating tool ... The paper would be better off having this methodological contribution as the main message rather than the separation of quantum-classical performance.*

Response: Thank you for this closing recommendation, which we have taken as the identity of the revised paper. The title leads with “Proxy-Based Diagnostics”; the abstract announces a proxy-based diagnostic and hypothesis-generation framework; the contribution list names the landscape as the paper’s central diagnostic and hypothesis-generation artifact; Section 7.4 states the falsifiable hypotheses it generates; and the Conclusions open by identifying the landscape as the primary contribution and hypothesis generation as the secondary one, followed by the sentence that the paper demonstrates no quantum advantage and implements no quantum algorithm.

We hope that the revised manuscript now meets the standards of *Sensors*, and we look forward to hearing from you in due course. We would be glad to respond to any further questions or comments.

Sincerely,

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